NIHR | National Institute for Health and Care Research

Reference number	NIHR208845
Programme	Research for Patient Benefit
Call	RfPB Competition 54 Stage 1
Lead Applicant	Dr Enrico De Luca
Host organisation	Royal Devon University Healthcare NHS Foundation Trust
Research Title	Co-designing training for care-home staff to improve patient relationships by understanding touch interactions
Start Date	01/07/2025
Grant Duration (Months)	24
End Date	30/06/2027
Estimated Research Cost	£195,589.11
Estimated NHS Support & Treatment costs	£444.00
Research Type	Primary Research

Lead Applicant Details

Lead Applicant

Full Name	Dr Enrico De Luca	**Position**	Senior Lecturer
ORCID iD	0000-0003-1516-3198	**Institution**	University of Exeter

Specify your (lead applicant) role in the research
Principal Investigator - responsible for the design and overall conduct of the research project: 1. The research protocol includes relevant literature, include PPIE insights, is scientifically sound, safe, ethical, legal and feasible. 2. Project development and management incorporates expert peer review and support. 3. Conducting and Coordinating Ethnographic study with research team and assistant 4. Ethical approvals are obtained and adhered to. 5. Support research team members 6. Budget is managed, reported and monitored. 7. Sound and robust conduct in recruitment of participants. 8. Data collection and analysis are adhered to and monitored. 9. Findings are effectively disseminated through multiple methods and mediums.

% FTE commitment	20

Research Team Details

Co-Applicant	**Organisation**	**Specify Co-Applicant role in research**	**% FTE**
Professor Louise Allan	University of Exeter	Professor Allan is a specialist interest in the Neurology and Psychiatry of Old Age. She will contribute to: • research design, • data analysis and collection, • dissemination	2.5
Professor Richard Kyle	University of Exeter	Professor Richard Kyle has held senior leadership positions in HE and NHS. He will contribute: • Research design • data analysis, • support development of training • support dissemination	2.5
Dr Holly Sugg	University of Exeter	Dr Sugg has expertise in developing/piloting/evaluating complex interventions. She will support and advise on: • training design and piloting • analysis for intervention development components.	1
Professor Karen Mattick	University of Exeter	Professor Mattick designed/managed research adopting realist research, scoping reviews, qualitative and mixed methods research studies. She will provide support/advice to: • research design • data collection/analysis • dissemination	1
Miss Faye Doris	University of Exeter	Miss Faye Doris is the PPIE Lead. She will: • Oversee PPI activity • Contribute to research design • Contribute to data collection and analysis • Contribute to dissemination	1
Mrs Tracey Hughes	BAY COURT RESIDENTIAL & NURSING HOME	Mrs Hughes is a care home manager, nurse and educator. She will contribute and facilitate: • Research design and setting • Care home support • Data collection	1

Co-Applicant	Organisation	Specify Co-Applicant role in research	% FTE
		• Dissemination	

COI declaration
Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.

Nothing to declare

Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

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Co-designing training for care-home staff to improve patient relationships by understanding touch interactions

The UK's aging population includes many older people with dementia who need to be compassionately and competently cared for in both care/nursing homes and at home. Meanwhile, the care-worker sector is plagued with recruitment and retention problems, and the wellbeing of (often international) staff is a serious issue. When care workers learn how to use a caring and gentle touch when interacting with people with dementia it significantly relieves distress, anxiety, and agitation, and has important benefits for the residents, staff and relatives.

Aim

This research will study how residents, relatives and staff feel about the way they touch and are touched during caring activities and daily interactions in care homes. Based on what we learn, we will work with the residents and staff to design appropriate touch training for care-workers and nurses.

Background

There are gaps in what we know about how people are affected by touch, especially for older people living with dementia. By carefully exploring touch interactions in care homes we will learn what staff know (and don't know) about the different types of touch and how they affect residents.

Methods

1.Scoping review

It consists in a literature review to map out What do (and don't) we know about how touch is used in the care of older people, dementia care, care homes and social care settings.

2.Ethnographic study in three care homes

An ethnographic approach means carefully observing people in their real-world activities, and using conversations and interviews to gain in-depth insight into how people interact in care homes.

3.Design training on "Awareness of interpersonal physical contact during caring encounters."

1&2 will give us the information to co-design training alongside care workers, customising it for care-homes and truly improving care through more meaningful touch.

Patient and public involvement (PPIE)

Exeter Academy of Nursing PPIE chair Faye Doris and member Howard Almond (himself a dementia carer) support this research. Our PPIE group of residents, relatives, and staff will ensure not only that study design and materials are appropriate, but also that all ethical and safeguarding issues are covered, including consent. PPIE members will feed back during the training development, and the final version will be presented during a half-day seminar to stakeholders and staff. This will help to develop future grants application for a feasibility study to roll out training on a larger scale.

Dissemination

In addition to publishing and presenting our results widely, our Advisory Board includes key people from NHS, Nursing, Dementia, and Care Home networks to ensure that the proposed new training will be enthusiastically and confidently rolled out without delay.

Research Plan

Research Plan

1. **What is the problem being addressed?**

This project responds to the current NIHR themed call on Dementia and the research priority 'complex health and care needs in older people'.

The UK has an aging population, and therefore a large number of older people with dementia who need to be compassionately and competently cared for in both care/nursing homes and at home. At the same time, the care-worker sector is plagued with recruitment and retention problems, and staff wellbeing is a serious issue.

Dementia and care homes. Nearly half a million people live in care homes in the UK[6,23] (70% residential, 29% in nursing homes). About 70 per cent of them have dementia or severe memory problems. By 2025, it's estimated that over 1 million people in the UK will have a diagnosis of dementia.[8] Good care homes try to provide the meaningful engagement and activities that improve health and wellbeing, but **residents can struggle with a perceived lack of meaning and purpose** – on top of the dementia symptoms.[3,9] Nurses and dementia experts observe how residents interact with their environment to understand which factors contribute to an individual's anxiety or aggression, and then try to prevent those situations.[18,36] International guidelines suggest nonpharmacological interventions like music therapy, light touch, and massage to prevent and treat agitation and delirium in dementia.[1,18]

Ageing and touch. In traditional healthcare settings, **the physical touch experienced by older individuals is often more task-oriented than caring**. Some older people – whether they live in a care home or are one of the 3.3 million people over 65 who live alone[25] – experience psychological and physiological conditions such as loneliness, isolation, neurological and cognitive impairment, and dementia.[2] The lack of social interactions and the absence of physical touch can exacerbate a person living with dementia's sense of isolation. Neuroscientists and experts have coined the term **"touch hunger"** to describe the referred need for meaningful interactions, especially in healthcare contexts, due to the lack of social touch for older individuals.[10] The lack of caring compassionate touch can increase anxiety and insecurity; diminish trust in caregivers; and decrease sensory awareness for medically frail older people.

Care homes staff. The relationship between staff and residents living with dementia and their relatives can be challenging.[15] **Care workers can lack confidence in dementia communication skills,**[22,24] and the tools to manage difficult communication, and this can lead to further frustration and intention to leave the job. Skills for Care highlighted the social care sector's high turnover (29%), and one factor is the lack of training or opportunities for development and new qualifications.[31] Lack of meaning in communication and person-centred care can result in staff compassion fatigue and burn out.[5] Care-home nurses and carers are frustrated by the stigma associated with social-care careers, and with being mostly an internationally recruited workforce with little support and lot of pressure.[6]

2. **Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients and health care services?**

The Alzheimer's Society emphasizes that research efforts should focus on improving the lives of people living with dementia today, not just finding a long-term cure.[4] Various governmental and local NHS initiatives are supporting the sector to deliver better person-centred care,[9,23,31] but workforce and funding crises are ongoing, exacerbated by the COVID-19 pandemic.[17] There is a real need for more effective care methods for people living with dementia, particularly ones that are easy for staff to learn. This research is important in improving the health and wellbeing of both patients and healthcare services because it will:

a. ***Extend and deepen our understanding of the importance of touch in dementia care.*** While research suggests that patients who receive caring touch experience less anxiety, physiological stress, and pain than patients who do not,[35] very few studies have explored specifically how awareness of interpersonal physical touch in caring encounters can improve communication with older people living with dementia and alleviate their feelings of loneliness and neglect in residential settings.[13,26]
b. ***Enrich evidence of how touch also influences the person initiating the touch.*** It is crucial to emphasise how touch also influences and can benefit those who initiate and have an active role in the tactile interaction. Touch is inevitable in care work, and documenting the positive impacts of touch and making it part of training will make care work more meaningful and rewarding, and may reduce personnel compassion fatigue, improving their wellbeing and increasing the chances of effective person-centred care.[13,26] The specific touch awareness acquired (or rediscovered) by health professionals is fundamental to understanding and advocating for when touch is inappropriate, or when it is the right tool to use for the right person and in the right setting.[7,27]
c. ***Lead to the co-creation of meaningful communication tools.*** Touch is a multi-sense phenomenon that communicates "beyond words"[16] (p. 206). Several frameworks that support, improve, and orientate care for people living with dementia use methodologies and approaches that include touch, sounds, and calming presence to create a therapeutic environment that promotes engagement, independence, and wellbeing. Touch can serve as a preparatory element for many interventions/caring activities and provide space-time reorientation, early mobilization, minimization of restraint use, and adequate sleep hygiene, which are all known to reduce agitation in dementia.[1,15,24]

3. **Review of existing evidence – How does the existing literature support this proposal?**

Touch studies. The sense of touch has inspired scientific theories in fields from developmental psychology to neuroscience and philosophy.[10] The tactile experience of touch, or a lack of it, affects the development of behaviour. Hertenstein et al. underlined how numerous behaviours are influenced by a physical contact experience – whether or not the experience of interpersonal contact is explicitly remembered.[14] Touch remains an important research subject, with a recent Nobel Prize awarded for discoveries related to receptors for temperature and touch.

Touch in Healthcare. Touch is an essential part of care activities, from providing fundamental personal hygiene and comfort to

performing clinical task/procedures. Touch affects our physiological and psychological well-being and an extensive body of research has explored the beneficial and therapeutic effects of physical contact in healthcare settings.[16] Nursing research highlights how touch can influence patients' wellbeing by reducing anxiety, facilitating communication and relationship, improving a sense of presence, and infusing security and encouragement. While cultural background, gender, and contextual factors can determine touching preferences, most studies also show that one's history or experience of touch (and traumas) influences how comfortable we are with touch. **Touching and learning how to touch is fundamental for health professions and carers**. Nurses who are comfortable with touch are less likely to experience compassion fatigue. Reflecting on interpersonal contact or using gentle touch techniques can reduce feelings of non-belonging and job dissatisfaction for nurses.[7,21]

Affective touch. Recent neurobiological evidence suggests humans have a specialised neural pathway to detect and process socially relevant touch.[10] Unlike touch to identify or manipulate objects (discriminative touch), socially relevant touch tends to be gentle, slow, and stroking (affective touch).[21] Stimulation of the fibres of the affectivity (called CT unmyelinated) afferents generates a pleasurable and rewarding experience because CT afferents activate social processing, emotional sensitivity, and brain reward areas.[10] There is some evidence that this neurobiological path is a necessary addition to the relational-cognitive path because receiving physical contact itself affects well-being even when not provided by a close friend or partner.[10,14] Moreover, distinct emotions and intentions can be communicated through touch. Affective touch studies confirmed slow, gentle, and moderate pressure touch communicates positive emotions and intentions.[19,21]

Affective touch and older people. Older people with dementia retain the ability to discern positive and caring intentions in physical contact such as touch and massage thanks to the retention by their limbic systems of the ability to decode emotions induced by touch.[21] In addition, while discriminative touch is negatively correlated with age, with increasing age the affective aspects of touch improve.[20] Therefore, tactile communication, especially affective touch, might be important in conveying meaning that is not communicated in any other way because it is based on a reciprocal pleasurable experience. Many experts recommend prioritizing non-pharmacological interventions to reduce agitation in patients with dementia, and systematic reviews have shown that sensory approaches such as touch and massage have moderate to high efficiency in managing/reducing agitation in this population.[28]

While the sense of touch is an ongoing research topic, there is a need for more studies focusing on understanding and improving physical contact between caregivers and older individuals with dementia in social care settings. Additionally, there is a lack of documented experiences related to touch education in healthcare professions and few conceptual/operational care models that consider intentional, mindful contact or a reflective approach to interpersonal touch.[34]

Therefore, this research seeks to fill such gaps by firstly exploring and understanding touch interactions that occur during care encounters in care homes and to shed light on the level of staff awareness, different type of touches adopted, and residents' responses. Secondly, it will lead to co-designing a pilot-training with a participatory approach by including care home residents, relatives and staff insights.

4. **What is the research question / aims and objectives?**

How are the touch interactions during caring activities and daily interactions in care homes perceived by residents, staff, and relatives?

Aims:

To observe and understand touch interactions between residents and their care workers in caring and daily activities in care homes.

To understand residents', staff and relatives' views and opinions on touch perceptions during caring and daily activities.

To design a blueprint for a touch-focused package of training for care-home workers.

Objectives:

Objective 1: To observe, adopting an ethnographic approach, touch interactions between residents and staff that occur during caring and daily activities in a care home.

Objective 2: To analyse the interpretations of physical contact and touch interactions in care activities that emerge from interviews with 3 care homes' residents, staff, and relatives.

Objective 3: To co-create an educational intervention focused on touch, tailored to care-home staff and residents' needs by co-participation design and practice development.

5. **Project plan**

The RfPB grant will support a research project involving care home residents, staff, and relatives. (The term 'care home' includes nursing and residential homes.)

The project will proceed in three phases:

Stage 1. Scoping review - Map available evidence on the use of interpersonal physical contact (including the diverse types of touch techniques) in clinical settings, and report on the types of evidence that address and inform practice in the field and the way the research has been conducted. Using the Joanna Briggs Institute framework for scoping reviews enhanced by Peters et al. (2020)[29] to highlight research knowledge gaps and refining the design of research questions, the review will focus on touch studies in the context of care of older people, dementia care, care homes, and primary care settings.

Stage 2. Ethnographic study of three Exeter-area care homes. The Ethnographic approach investigates specific beliefs and practices of healthcare processes and specific clinical settings, as managed by patients and practitioners.[12] Study participants will be the care-home residents, staff and relatives. The ethnographic study will include immersive observations, review of practice documents and protocols, informal conversations with staff, and semi-structured interviews with staff and residents. The observations will take place weekly (4 hours) for a period of six months in each care home. The researchers will write field notes to describe participants' interactions, details of the contexts in which the action takes place, and sensory impressions. Residents, family members and staff unstructured interviews will be conducted and audio recorded. The residents' interviews will be conducted carefully using gentle and creative methods tailored for people with dementia and cognitive impairment.[33] The ethnographic data analysis is an iterative process that starts during fieldwork and includes reflection, which is fundamental to value the different interpretations emerging during the analysis itself.

Stage 3. Design of a training on "Awareness of Interpersonal physical contact during caring encounters." Training design will be guided by the scoping review and ethnographic study. Ethnographic methods ensure person-centred design and customized interventions for effective implementation in specific contexts. An experience-based co-design approach will effectively and meaningfully engage care-home staff in the organic development of the training, ensuring it is tailored to their specific needs/setting and guided by their expert lived experience. Patient and Public Involvement and Engagement (PPIE) members and stakeholders will provide feedback during the training development, and the final version will be presented during a half-day seminar to stakeholders and staff. This will help to develop future grant applications for a feasibility study to roll out training on a larger scale.

Study Participants

The PI has established a relationship with a care homes network, securing the participation of three care homes in South Devon. South Devon is one of the most popular retirement choices in the UK and has a high density of care homes. A preliminary presentation of the project to care-home managers in May 2024 resulted in fruitful conversation and practical suggestions to set up the research, communicate effectively, and safely and respectfully recruit participants (residents, care-home staff, and relatives).

The study will invite all residents to participate. During the ethnographic data collection, researchers will plan, inform, and seek permission to observe caring activities and touch interactions. The consent to participate of people with cognitive impairment and reduced ability will be facilitated by relatives or trusted carers, who will be supported to create the right environment to conduct unstructured interviews. Residents, families, and staff do not have to participate if they do not wish to do so.

The PI is an experienced nurse, and he will provide a safe and trustworthy environment for the study. Discussion has already begun with care-home managers regarding shaping an ad hoc honorary contract for this project.

Ethical considerations

The Social Care Research Ethics Committee review of the study will be obtained via Integrated Research Application System. The ethical process and approval will be crucial for this project, as working in such a delicate setting (care homes) will require several steps at the very beginning of the project. The PPIE group will help shape research tools and adapt them for the specific care setting.

Researchers will ensure that residents, staff, and relatives are informed of and understand ethnographic observation and data collection, both verbally and by using visual aids like small posters and leaflets. The study will be inclusive of people who lack capacity or lose capacity to provide consent, in keeping with the EDI view that a lack of capacity to consent should not be grounds for exclusion by the team. Researchers will align with the care home measures to safeguard residents by attending the specific training.

PPIE

The project has the support of PPIE group members from the Exeter Academy of Nursing. The PPIE chair (Miss Faye Doris) and group member Mr Howard Almond have reviewed this application and pledged to support this research. Mr Almond has direct experience of caring for a person with dementia, which will provide invaluable insight and enrich this work. The PI presented the project to an NIHR Peninsula Public Engagement Group virtual ideas café, gaining valuable feedback.

Residents, their family members/significant others, and care-home workers (one representative from each home) will be invited to join the project PPIE group. They will advise and guide study design, recruitment, and dissemination of materials and activities. They will also review participant information leaflets and consent forms and suggest ways to disseminate findings.

The bespoke PPIE group will meet for one hour at key stages of the project, such as protocol review, recruitment, data collection, and dissemination, up to eight times. Members will be remunerated in accordance with NIHR guidance.

The PPIE group will also ensure the ethical challenges of setting up a project in such a specific environment are met. In particular, the PPIE member in the care-home environment will assist with safeguarding issues, such as inappropriate touch, the use of relatives to safeguard residents who are unable to provide consent, and any concerns related to the safe and respectful involvement of residents.

Equality, Diversity, and Inclusion

As PI I will use my knowledge and experience as departmental EDI lead to ensure that the principles of equality, diversity, and inclusion are respected throughout the project.

Residents: As the research area is not very ethnically diverse, we will focus on differences in social background and class among residents to enhance participant diversity, e.g. the ratio between privately/local authority-funded residents. The chosen care homes include both rural and urban populations, and one is in a deprived area of South Devon.

Care home staff: The care home workforce, predominantly international staff, plays a crucial role in enriching the research landscape. Their diverse perspectives, cultural backgrounds, and experiences regarding touch in care are invaluable contributions to the research on cultural and ethnic diversity.

Research material, including posters, leaflets, and information sheets, will be carefully adapted and tested to meet the cultural and cognitive needs of the residents and their relatives.

Dissemination

The project's key learnings and best practices will be shared with the broader health and social care sector through stakeholders and agencies, journal articles, conference presentations, public engagement events, social media, and digital platforms. This will include local, national, and international impact in care homes and HEIs.

Advisory board

The Advisory Board will meet four times throughout the 2-year project. Members include NHS England (Jane Bunce), NHS Devon Integrated Care Board (Stevie Morris), Chief Nurse for Social Care (Debra Sturdy), Skills for Care Project Manager (Lucy Gillespie), Devon Training Hub-Primary Care (TBC), Dementia UK (TBC). These members, including health economist Prof Hulme and neuroscientist Prof Cazzato (Liverpool University) will actively contribute to the study's development at all stages, from design to implementation to knowledge exchange. The PPIE lead will facilitate communication between the PPIE and advisory groups.

Timetable (Months):

1-6	• Scoping review • RA recruitment • Ethics application • Care homes and study participants recruitment • Commence and schedule PPIE/Advisory board/stakeholders meetings
6-18	• Setting up ethnographic study. • Data collection (6 months) - observations and interviews • Data analysis • Sharing preliminary and final findings with study participants, PPIE and stakeholders
18-24	• Co-designing pilot training • Workshop/presentation events. • Writing publications and final report. • Dissemination activities

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Uploads

Reference List

1. Abraha, I., Rimland, J. M., Trotta, F. M., Dell'Aquila, G., Cruz-Jentoft, A., Petrovic, M., ... & Cherubini, A. (2017). Systematic review of systematic reviews of non-pharmacological interventions to treat behavioural disturbances in older patients with dementia. The SENATOR-OnTop series. BMJ open, 7(3), e012759.
2. Age UK (2023) Loneliness research and impact | Age UK consulted 17th October 17, 2023
3. Alzheimer Society (2023) What is Dementia? https://www.alzheimers.org.uk/about-dementia/types-dementia/what-is-dementia
4. Alzheimer Society (2024) Dementia research roadmap for prevention, diagnosis, intervention and care by 2025. https://www.alzheimers.org.uk/research/for-researchers/research-roadmap
5. Costello, H., Cooper, C., Marston, L., & Livingston, G. (2020). Burnout in UK care home staff and its effect on staff turnover: MARQUE English national care home longitudinal survey. Age and Ageing, 49(1), 74-81.
6. Day, J., Dean, S. G., Reed, N., Hazell, J., & Lang, I. (2022). Knowledge needs and use in long-term care homes for older people: A qualitative interview study of managers' views. Health & Social Care in the Community, 30(2), 592-601.
7. De Luca, E., Fatigante, M., Zucchermaglio, C., & Alby, F. (2021). "Awareness to touch": A qualitative study of nurses' perceptions of interpersonal professional contact after an experiential training. Nurse Education in Practice Volume 56, October 2021.
8. Dementia UK (2024) What is dementia?. https://www.dementiauk.org/information-and-support/about-dementia/what-is-dementia/ consulted 30/06/24.
9. Dept of Health and Social Care (2023). Chief Medical Officer's annual report 2023: health in an ageing society. Accessed online 23 April 2024 https://www.gov.uk/government/publications/chief-medical-officers-annual-report-2023-health-in-an-ageing-society
10. Gallace, A. & Spence C. (2010). The science of interpersonal touch: An overview. *Neuroscience & Biobehavioral Reviews*, 34, 246–259
11. Haines et al. (2022) Ethnography and user-centered design to inform context-driven implementation, Translational Behavioral Medicine, Volume 12, Issue 1, January 2022, ibab077, https://doi.org/10.1093/tbm/ibab077
12. Hammersley, M. & Atkinson, P. (2019). Ethnography: Principles in practice. Routledge.
13. Heatley-Tejada, A., Dunbar, R. I. M. & Montero, M. Physical contact and loneliness: being touched reduces perceptions of loneliness. Adapt. Hum. Behav. Physiol. 6, 292–306 (2020).
14. Hertenstein et al. (2006) The communication of emotion via touch. Emotion, 9(4), 566.
15. Höglander J, Holmström IK, Gustafsson T, Lindberg E, Söderholm HM, Hedén L, von Heideken Wågert P, Sundler AJ. Implementing A person-centred CommunicaTION (ACTION) educational intervention for in-home nursing assistants - a study protocol. BMC Geriatr. 2023 Feb 25;23(1):112. doi: 10.1186/s12877-023-03831-3. PMID: 36841761; PMCID: PMC9960173.
16. Kelly, M. A., Nixon, L., McClurg, C., Scherpbier, A., King, N., & Dornan, T. (2018). Experience of touch in healthcare: A meta-ethnography across the healthcare professions. *Qualitative Health Research,28*(2), 200–212.
17. King's Fund (2024) 'The adult social care workforce in a nutshell' https://www.kingsfund.org.uk/insight-and-analysis/data-and-charts/social-care-workforce-nutshell consulted 01/07/2024
18. Livingston, G., Huntley, J., Sommerlad, A., Ames, D., Ballard, C., Banerjee, S., ... & Mukadam, N. (2020). Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. The Lancet, 396(10248), 413-446.
19. McGlone, F., Wessberg, J., & Olausson, H. (2014). Discriminative and affective touch: sensing and feeling. *Neuron*, *82*(4), 737-755.
20. McIntyre S, Nagi SS, McGlone F, Olausson H. The Effects of Ageing on Tactile Function in Humans. Neuroscience. 2021 Jun 1;464:53-58. doi: 10.1016/j.neuroscience.2021.02.015. Epub 2021 Feb 17. PMID: 33607227.
21. Morrison & Croy, I. (2021). The Science of Social and Affective Touch. Neuroscience, 464, 1-2.
22. Nguyen, H., Terry, D., Phan, H., Vickers, J., & McInerney, F. (2019). Communication training and its effects on carer and care-receiver outcomes in dementia settings: A systematic review. Journal of Clinical Nursing, 28(7-8), 1050-1069.
23. NHS England (2023). Enhanced health in care homes framework. https://www.england.nhs.uk/publication/enhanced-health-in-care-homes-framework/ consulted 30/06/24
24. O'Brien, R., Goldberg, S. E., Pilnick, A., Beeke, S., Schneider, J., Sartain, K., ... & Harwood, R. H. (2018). The VOICE study–A before and after study of a dementia communication skills training course. PloS one, 13(6), e0198567.
25. Office of National Statistics (ONS). https://www.ons.gov.uk/census Consulted 14th October 2023
26. Packheiser, J., Hartmann, H., Fredriksen, K. et al. A systematic review and multivariate meta-analysis of the physical and mental health benefits of touch interventions. Nat Hum Behav (2024). https://doi.org/10.1038/s41562-024-01841-8
27. Pedrazza, M., Berlanda, S., Trifiletti, E., & Minuzzo, S. (2018). Variables of individual difference and the experience of touch in nursing. Western Journal of Nursing Research, 40(11), 1614-163
28. Perini, I., Olausson, H., & Morrison, I. (2015). Seeking pleasant touch: neural correlates of behavioral preferences for skin stroking. Frontiers in behavioral neuroscience, 9, 8.
29. Peters, MDJ, Godfrey C, McInerney P, Munn Z, Tricco AC, Khalil, H. Chapter 11: Scoping Reviews (2020 version). In: Aromataris E, Munn Z (Editors) JBI, 2020. Available JBI Manual for Evidence Synthesis from . https://synthesismanual.jbi.global https://doi.org/10.46658/JBIMES-20-12
30. Pinto et al., 2020 Pinto, J. O., Dores, A. R., Geraldo, A., Peixoto, B., & Barbosa, F. (2020). Sensory stimulation programs in dementia: A systematic review of methods and effectiveness. Expert Review of Neurotherapeutics, 20(12), 1229-1247.
31. Skills for Care (2023) The state of the adult social care sector and workforce in England. www.skillsforcare.org.uk/stateof
32. Sugg, H. V. R., Richards, D. A., & Frost, J. (2017). Optimising the acceptability and feasibility of novel complex interventions: an iterative, person-based approach to developing the UK Morita therapy outpatient protocol. Pilot and Feasibility Studies, 3, 1-11.
33. Tischler, V., Zeilig, H., O'Malley, M., & Asker, C. (2023). Together yet apart: Rethinking creativity and relational dementia care during the Covid-19 pandemic. Geriatric Nursing, 54, 99–107. https://doi.org/10.1016/j.gerinurse.2023.08.012
34. Wilson, M., Gettel, V., Walsh, J., & Esquenazi, S. (2016). Caring compassionately for hospitalized patients: Can nurse-delivered massage address compassion fatigue?. International Journal for Human Caring, 20(3).
35. Wu, J., Wang, Y., & Wang, Z. (2017). The effectiveness of massage and touch on behavioural and psychological symptoms of dementia: A quantitative systematic review and meta-analysis. Journal of advanced nursing, 73(10), 2283-2295.legeforsch. 2018;4:13–26. doi: 10.6094/KlinPfleg.4.13.
36. Zaninotto, P., Iob, E., Demakakos, P., & Steptoe, A. (2022). Immediate and longer-term changes in the mental health and well-being of older adults in England during the COVID-19 pandemic. JAMA psychiatry, 79(2), 151-159.

Project Gantt Chart

	Project preparation			24 Months																							
Activities	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Protocol development																											
Preparing study documentation																											
Research team meetings																											
PPIE meetings																											
Advisory board meetings																											
Scoping review																											
Recruitment RA																											
Selecting the setting																											
Recruitment of study participants, collaborators and resources for ethnographic study																											
Ethic application and approval																											
Data collection (Field work and Qualitative interviews)																											
Data analysis																											
Member checks and discussion of preliminary findings																											
Findings report and discussion																											
Co-design activities and training development																											
Workshops for training presentation to stakeholders and staff																											
Writing for publications																											
Preparing dissemination materials and conference abstract submission.																											
Conference participation																											
Dissemination activities																											
Final Report Writing																											